Synthetic Fixed-Confidence Experiments for Binary Pairwise Pareto Set Identification

Experiment report generated from `Exp_Synthetic`

July 18, 2026

Abstract

This report summarizes the completed synthetic experiments for fixed-confidence binary pairwise Pareto set identification. The experiments compare VB-EGE-practical against three fixed-confidence benchmarks: focal Borda, BT-MLE-Cert, and pairwise Borda plug-in. The MLE method uses a fixed-confidence-style empirical certificate and is not claimed to have a formal delta-correct confidence region. The main runs contain 2400 scaling replications and 3200 benchmark replications per algorithm across all experiment cells. Across all completed main settings, the empirical error rate is 0.0 for scaling and 0.0 for benchmarks; therefore the main comparison is sample complexity. Mean stopping time with its cluster-aware standard error is the primary summary throughout; paired mean slowdown ratios and tail quantiles provide complementary diagnostics. The experiments are organized in logical order: protocol and empirical error evidence, constant sensitivity, fixed-confidence scaling, confidence-scaling quantiles, the fixed-confidence benchmark suite, Pareto-size ablation, and sampling-mechanism sanity checks.

1 Protocol and Empirical Error Evidence

The task is fixed-confidence identification of the strict Pareto set under binary pairwise comparison feedback. Each run stops adaptively when the algorithm can return a Pareto set certificate at the target failure probability. The default target is $\delta = 0.05$, except in the confidence-scaling sweep where δ itself is varied. The reported stopping time counts coordinate-specific pairwise comparison pulls. Lower stopping time is better when the empirical error rate is controlled.

Empirical error is the fraction of runs whose returned set differs from the true Pareto set. All completed main fixed-confidence cells have zero observed errors. A zero count is summarized by the 95% Wilson upper bound rather than interpreted as proof of zero risk. For example, 500 zero-error replications give a Wilson upper bound of about 0.0076, so the $\delta = 0.001$ sweep supports the stopping-time trend in $\log(1/\delta)$ but cannot directly validate a one-in-a-thousand error rate.

Table 1: Algorithms compared with fixed-confidence stopping interfaces. The MLE row is an empirical certificate heuristic, as detailed below.

Name	Sampling model	Fixed-confidence rule
VB-EGE-practical	Coordinate-wise Vector-Borda estimates	Adaptive accept/reject using empirical gap certificates
UniformFocalBorda-FC	Uniform coordinate-wise Borda sampling	Same confidence threshold family without adaptive elimination
UniformPairwiseBT-MLE-Cert	Uniform pair-coordinate Bradley–Terry sampling with MLE	Empirical plug-in certificate; no formal MLE confidence region
UniformPairwiseBT-BordaPlugIn-FC	Uniform pair-coordinate sampling with Borda plug-in	Stop when Borda plug-in certificate is satisfied

Definitions used by the report. There are K arms and d objectives. A coordinate-specific comparison between arms i and j on objective r follows a Bradley–Terry model,

$$Y_{ijr} \sim \text{Bernoulli}(p_{ijr}), \qquad p_{ijr} = \sigma(\theta_{ir} - \theta_{jr}).$$

The coordinate-wise Borda embedding is

$$B_{ir} = \frac{1}{K-1} \sum_{j \neq i} p_{ijr}.$$

Most certificates in this report are computed in this Vector-Borda space. The synthetic generators used for the main experiments pass a Borda/latent Pareto consistency check, so the reported true Pareto set agrees under θ and under B.

For any score matrix $X \in \mathbb{R}^{K \times d}$, define

$$m_{ij}(X) = \min_r (X_{jr} - X_{ir}), \qquad M_{ij}(X) = \max_r (X_{ir} - X_{jr}), \qquad \Delta_i^\star(X) = \max_{j \neq i} m_{ij}(X).$$

If arm i is non-Pareto, its identification gap is $\Delta_i(X) = \Delta_i^\star(X)$. If arm i is Pareto, the reported gap is

$$\Delta_i(X) = \min_{j \neq i} \min\{M_{ij}(X), \max(M_{ji}(X), 0) + \max(\Delta_j^\star(X), 0)\}.$$

The minimum Borda gap is $\Delta_{\min,B} = \min_i \Delta_i(B)$. The Borda hardness is

$$H_B = \sum_i \Delta_i(B)^{-2},$$

with $H_B = \infty$ when a gap is non-positive or non-finite. Large H_B means that one or more arms lie close to the Pareto decision boundary.

All fixed-confidence methods use the same practical constants: $c_{\text{samp}} = 2$, $c_{\text{thr}} = 4$, and $c_{\log} = 4$. In phase m, with cell count C, the schedule is

$$\varepsilon_m = 2^{-m}, \qquad L_m = \log(c_{\log} C m^2/\delta), \qquad n_m = \left\lceil c_{\text{samp}} L_m / \varepsilon_m^2 \right\rceil, \qquad r_m = \sqrt{L_m/(2n_m)}.$$

The sample constant c_{samp} multiplies the per-cell sample count, the log constant $c_{\log}$ is the union-bound safety factor inside L_m, and the threshold constant c_{thr} is the certificate margin. A fixed-confidence baseline stops when its estimated minimum gap satisfies $\widehat{\Delta}_{\min} > c_{\text{thr}} r_m$.

Practical versus theory constants. The proof-oriented schedule uses larger universal constants to preserve a worst-case concentration argument uniformly over arms, objectives, and phases; the explicit theory-style sanity check uses $(8, 16, 4)$. Our default $(c_{\mathrm{samp}}, c_{\mathrm{thr}}, c_{\mathrm{log}}) = (2, 4, 4)$ is therefore less conservative than the theory-style setting rather than numerically identical to it. This is reasonable for the finite synthetic regime because it keeps the same phase schedule and $\log(1/\delta)$ dependence, uses one fixed choice for VB-EGE and the comparable stopping certificates, and was checked by the constant-sensitivity and theory-constant experiments. The choice is not retuned by benchmark or algorithm. Except for the sections explicitly varying constants, every scaling, confidence, benchmark, and Pareto-size experiment uses this same practical choice.

Benchmark implementations. The three benchmarks other than VB-EGE are implemented as follows.

- **UniformFocalBorda-FC.** In each phase it samples every arm-objective cell (i, r) up to n_m Borda observations. Equivalently, the simulation draws $S_{ir} \sim \mathrm{Binomial}(N_{ir}, B_{ir})$ and estimates $\widehat{B}_{ir} = S_{ir}/N_{ir}$. It computes the plug-in strict Pareto set from $\widehat{B}$, computes $\widehat{\Delta}_{\min}$ using the gap formula above, and stops when $\widehat{\Delta}_{\min} > c_{\mathrm{thr}} r_m$. It does not eliminate arms adaptively; it keeps all Kd cells balanced.
- **UniformPairwiseBT-MLE-Cert.** In each phase it samples every pair-coordinate cell (i, j, r), $i < j$, up to n_m comparisons, so $C = dK(K-1)/2$. For each objective r, it fits a separate ridge-stabilized BT MLE from the win counts:

$$\min_{\vartheta_{\cdot r}} \sum_{i<j} \{W_{ijr} \log(1 + \exp[-(\vartheta_{ir} - \vartheta_{jr})]) + W_{jir} \log(1 + \exp[\vartheta_{ir} - \vartheta_{jr}])\} + \frac{\lambda}{2} \|\vartheta_{\cdot r}\|_2^2.$$

  The fitted coordinate scores are centered for identifiability. The algorithm then computes the plug-in Pareto set and $\widehat{\Delta}_{\min}$ from $\widehat{\theta}$ and stops when the same gap certificate is satisfied. If the graph is disconnected, the optimizer fails, or the estimate is too large, the implementation reruns the fit with the fallback ridge value. This method is labeled "Cert" rather than "FC" because the Bernoulli cell radius r_m is not, by itself, a valid uniform confidence radius for $\|\widehat{\theta} - \theta\|_\infty$. A formal BT-MLE guarantee would also need likelihood curvature, the comparison-graph Laplacian, dynamic range, ridge, and gauge fixing. Thus this baseline has a fixed-confidence-style stopping interface but is an empirical certificate heuristic; zero observed error is not presented as a proof of δ-correctness.
- **UniformPairwiseBT-BordaPlugIn-FC.** It uses the same balanced pair-coordinate sampling as the MLE baseline, but avoids solving the BT MLE. It estimates each pairwise probability with add-1/2 smoothing,

$$\widehat{p}_{ijr} = \frac{W_{ijr} + 1/2}{N_{ijr} + 1},$$

  forms the plug-in Borda estimate $\widehat{B}_{ir} = (K-1)^{-1} \sum_{j \neq i} \widehat{p}_{ijr}$, and applies the same strict-Pareto plug-in set and gap certificate in Borda space.

In the benchmark runs, the BT-MLE baseline uses balanced random pair-coordinate allocation, ridge parameter 10^{-8}, fallback ridge parameter 10^{-4}, maximum absolute latent score 20, and MLE tolerance 10^{-9} in the benchmark run. Raw outputs are stored as CSV files because the local parquet engine was unavailable; this does not change the numerical results.

Metrics and legends. Mean stopping time $\bar{\tau}$ with its standard error is the primary stopping-time summary in every section. The heterogeneous benchmark suite additionally reports paired per-replication mean slowdown ratios with cluster-aware standard errors. Across the main fixed-confidence runs the empirical Pareto-set error is zero; this is reported with Wilson upper bounds rather than uninformative zero-error floor curves. The formal baselines in the report are the fixed-confidence baselines, not capped fixed-budget variants. Plot legends use the short labels VB-EGE, Focal Borda, BT-MLE, and Borda plug-in. In line plots these are encoded respectively by solid circles, dashed open squares, dash-dot triangles, and solid diamonds, so nearly coincident curves remain distinguishable without relying only on color. Every uncertainty interval is drawn as a translucent shaded region: a ribbon around a line or a darker interval block over a bar. No whisker-style error bars are used.

Plot computation conventions. Whenever the same axis label appears later, it uses the same computation. Mean stopping time is $\bar{\tau} = n^{-1}\sum_s \tau_s$ over replications. For every plot whose center is a mean, the shaded uncertainty band is one standard error, $[\bar{\tau} - \widehat{\mathrm{SE}}(\bar{\tau}), \bar{\tau} + \widehat{\mathrm{SE}}(\bar{\tau})]$. When an instance has multiple observation replications, $\widehat{\mathrm{SE}}$ is computed from the latent-instance means and the effective sample size is the number of instances; otherwise it is the ordinary replication standard error. Every stopping-time curve and bar uses this mean $\pm$ SE interval as its shaded region. The q95 stopping time retained in the confidence section is the empirical 95th percentile of the same $\{\tau_s\}$ values. When an interval is narrower than the plotted line width, its ribbon visually coincides with the center line and is not artificially enlarged. In particular, several dyadic scaling baselines have exactly zero within-cell stopping-time variance, so their SE ribbon has zero width; the nonzero VB-EGE ribbons can also be visually narrow on a log axis. Empirical error is $n^{-1}\sum_s \mathbf{1}\{\widehat{P}_s \neq P_s\}$; when no failures are observed, error-floor plots show $\log_{10}(0.5/n)$ instead of $-\infty$. The Wilson upper bound is the upper endpoint of the two-sided 95% Wilson interval for this binomial error rate. A main-benchmark paired ratio first computes $R_s = \tau_{\mathrm{baseline},s}/\tau_{\mathrm{VB},s}$ on the same instance and observation replicate, then reports $\bar{R} = n^{-1}\sum_s R_s$ with one standard error. Where hierarchical instance IDs are available, that standard error is computed from latent-instance means; legacy fixed-instance cells use the ordinary replication standard error. Other ratio plots use the explicitly stated ratio of summary statistics. Pair-cell coverage is averaged over runs after computing τ_s/C for each pairwise run, where $C = dK(K-1)/2$. Mean Hamming distance is the average symmetric-difference size $|\widehat{P}_s \triangle P_s|$. Final phase is the last dyadic phase index m used by the algorithm.

2 Constant Sensitivity

Definitions. The sample constant c_s multiplies the phase sample count, the threshold constant c_θ scales the certificate margin, and $c_{\log}$ scales the logarithmic safety term. The practical choice is $(c_s, c_\theta, c_{\log}) = (2, 4, 4)$; the explicit theory-style check uses $(8, 16, 4)$. Heatmap colors encode mean stopping time and each cell prints mean $\pm$ one standard error. In the two bar panels, the darker overlay also spans mean $\pm$ one standard error.

Table 2: Arena-10 cross-section audit. The benchmark and constant-sensitivity rows use shared bank `arena10_medium_v2`: 30 fixed θ instances with 10 observation replications each and common random numbers. The confidence row retains its original paired-δ bank.

Section	Reps	Mean τ (SE)
Main benchmark	300	6.13×10^8 (6.46×10^6)
Confidence sweep, $\delta = 0.05$	300	6.21×10^8 (3.63×10^6)
Constant sensitivity, $(2, 4)$	300	6.13×10^8 (6.46×10^6)

The earlier cross-section discrepancy came from independent random instance banks: rare near-boundary Arena instances created very large stopping-time outliers, so raw means differed by orders of magnitude. The revised benchmark and constant-sensitivity rows now use the same 30 instances, ten observation replications, and common random numbers, so their default results agree exactly. The confidence sweep keeps its original independent bank because its within-replication pairing across δ is the design needed for the confidence-scaling claim; its mean remains informative within that paired bank but is not treated as an exact cross-section identity with the shared benchmark bank.

Table 3: Constant-sensitivity settings. VB-EGE is swept over sample constants $\{0.5, 1, 2, 4\}$ and threshold constants $\{2, 3, 4, 6\}$ at $\delta = 0.05$; the focal and BT-MLE-Cert references use the default constants $(2, 4)$.

Setting	Generator	Parameters	Grid	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	300
Witness-10	unique_witness_d	K=80; d=10; s=16; q=4; margins [0.035,0.14]	$c_s \in \{0.5, 1, 2, 4\}$; $c_\theta \in \{2, 3, 4, 6\}$	300

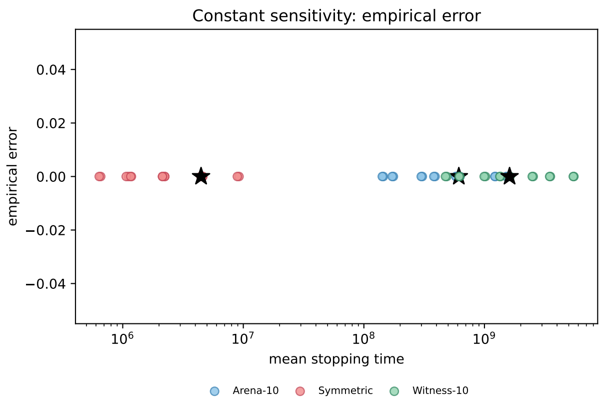
Table 4: Constant-sensitivity stopping times. The default row uses $(c_s, c_\theta) = (2, 4)$. The fastest row minimizes mean stopping time among grid points with empirical error at most 0.05 and complete stopping. Parentheses contain one standard error.

Setting	Default VB mean τ (SE)	Fastest constants	Fastest mean τ (SE)
Symmetric K64 d10	4.46×10^6 (0)	(4, 2)	6.38×10^5 (5724.38)
Arena-10 medium	6.13×10^8 (6.46×10^6)	(4, 2)	1.42×10^8 (8.86×10^5)
Witness-10	1.62×10^9 (4.99×10^6)	(2, 2)	4.75×10^8 (2.96×10^6)

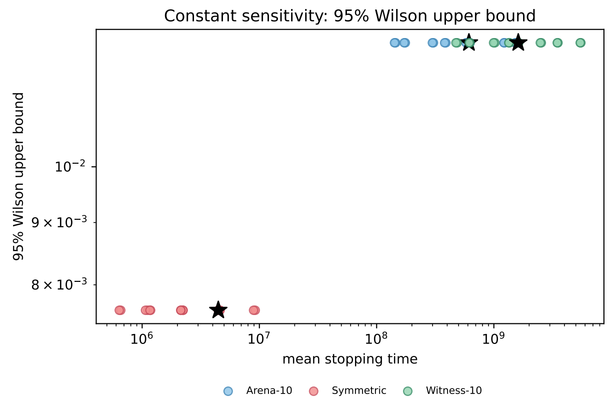
Table 5: Constant-sensitivity baseline ratios at the default practical constants. Ratios compare mean stopping times; values above one favor VB-EGE. Every tested constant pair has zero observed error, so the sweep is a sensitivity analysis rather than an identified efficiency–reliability frontier.

Setting	Focal/default VB	MLE/default VB
Symmetric K64 d10	1.00x	2.27x
Arena-10 medium	2.05x	4.82x
Witness-10	2.16x	2.94x

The sensitivity sweep varies the VB-EGE sample constant c_s and threshold constant c_θ. In the sensitivity figures, each point is one constant pair; the x-axis is mean stopping time and the y-axis is either empirical error or Wilson upper confidence bound. Heatmaps use sample constant on one axis and threshold constant on the other; darker or lighter cells encode the plotted metric named in the title. The default paper setting is $(c_s, c_\theta) = (2, 4)$.

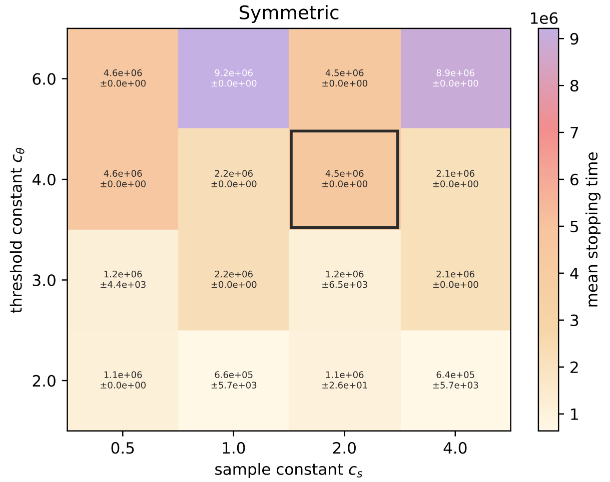


(a) Empirical error frontier

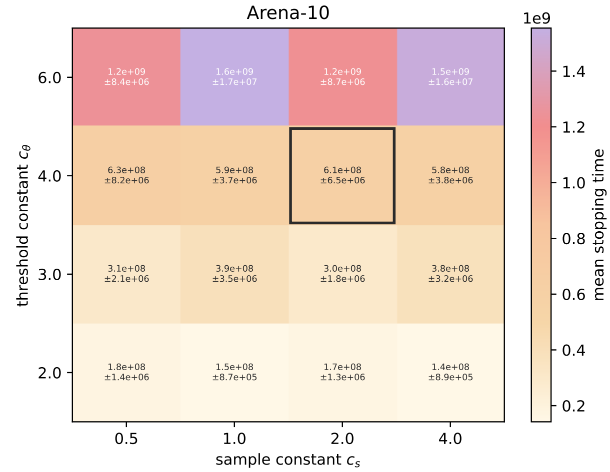


(b) Wilson upper frontier

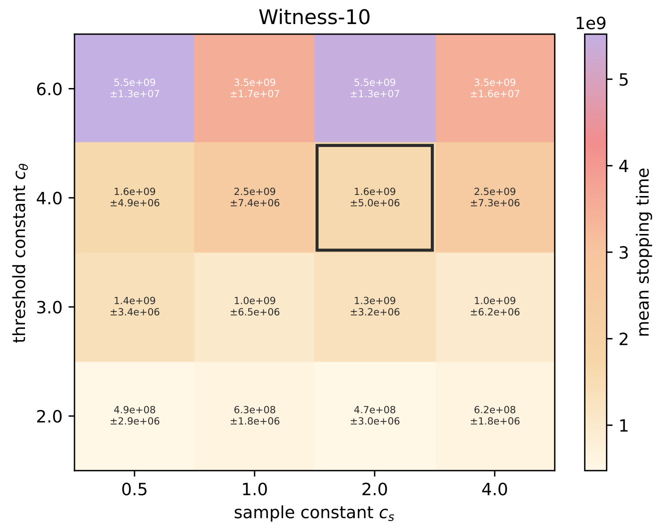
Figure 1: Constant-sensitivity error diagnostics. All tested cells have zero observed error; panel (b) shows the corresponding Wilson upper bounds.



(a) Symmetric K64 d10



(b) Arena-10 medium



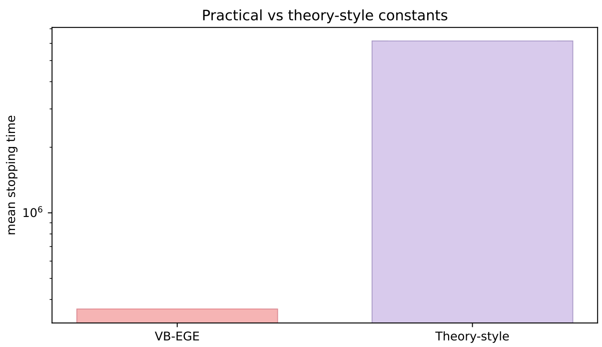
(c) Witness-10

Figure 2: Constant-sensitivity mean stopping-time heatmaps. Color encodes mean stopping time; cell text reports mean $\pm$ one standard error.

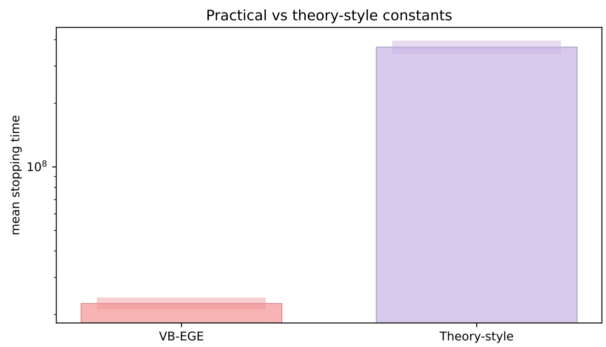
Result interpretation. This grid establishes stopping-time sensitivity, not a full efficiency-reliability frontier, because all observed errors are zero. Smaller constants reduce stopping time on these instances, while the default practical constants remain the common reference point for the main comparisons.

Table 6: Theory/practical constants sanity check. Both variants use the same instances and observation replications. The theory-style constants are $(c_s, c_\theta, c_{\log}) = (8, 16, 4)$ and the practical constants are $(2, 4, 4)$.

Setting	Practical mean τ (SE)	Theory mean τ (SE)	Theory/practical
Symmetric K16 d4	3.62×10^5 (1896.03)	6.15×10^6 (6471.04)	17.0x
Arena-4 K16	2.26×10^7 (1.45×10^6)	3.69×10^8 (2.62×10^7)	16.3x



(a) Symmetric K16 d4



(b) Arena-4 K16

Figure 3: Practical and theory-style constants on small instances. Darker overlays span mean ± one standard error; the theory-style constants are deliberately more conservative.

3 Fixed-Confidence Scaling

Definitions. This section studies instance scaling under a fixed-confidence stopping rule: the target failure probability is held fixed at $\delta = 0.05$ throughout. The three sweeps vary the number of arms K, objective dimension d, or latent gap Δ, holding the remaining instance parameters fixed. Dependence on δ is studied only in the next section, Confidence-Scaling Quantiles. Every line and bar reports mean stopping time $\bar{\tau}$. Its translucent ribbon or darker bar overlay is $\bar{\tau} \pm \widehat{\mathrm{SE}}(\bar{\tau})$ as defined in Section 1. Axes are logarithmic when all plotted values are positive; lower values use fewer pulls.

Table 7: Fixed-confidence scaling settings. All tasks use the symmetric hard generator and four algorithms; each cell is averaged over 200 random permutations/seeds.

Sweep	Generator	Grid	Purpose
K scaling	symmetric_hard	K in {16,32,64,128}; d=4; Delta=1; delta=0.05	One Pareto arm at the origin and K-1 dominated arms at -Delta in every coordinate.
d scaling	symmetric_hard	K=64; d in {2,4,10}; Delta=1; delta=0.05	Symmetric hard instance with the number of dimensions varied.
Gap scaling	symmetric_hard	K=64; d=10; Delta in {0.5,0.75,1,1.25,1.5}; delta=0.05	Symmetric hard instance with the latent separation Delta varied.

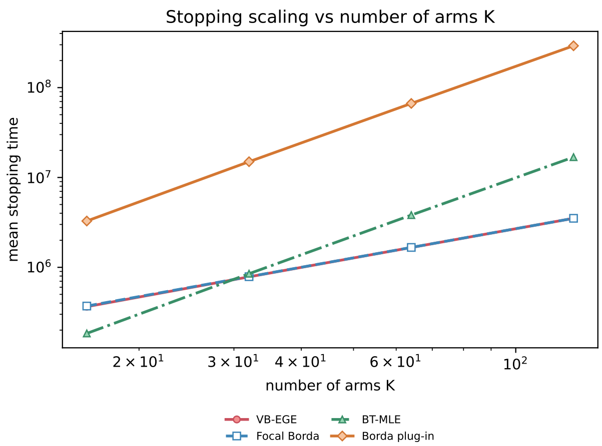
Table 8: Scaling means and baseline ratios. Parentheses contain one standard error for VB-EGE; all empirical error rates are zero.

Value	VB mean τ (SE)	Focal/VB	MLE/VB	Plug-in/VB
K scaling				
16	3.65×10^5 (838.12)	1.02x	0.50x	8.97x
32	7.84×10^5 (819.92)	1.00x	1.08x	19.1x
64	1.66×10^6 (989.48)	1.00x	2.29x	40.1x
128	3.51×10^6 (579.76)	1.00x	4.80x	83.2x
d scaling				
2	7.67×10^5 (2607.88)	1.03x	2.37x	41.6x
4	1.66×10^6 (542.58)	1.00x	2.29x	40.1x
10	4.46×10^6 (0)	1.00x	2.27x	39.5x

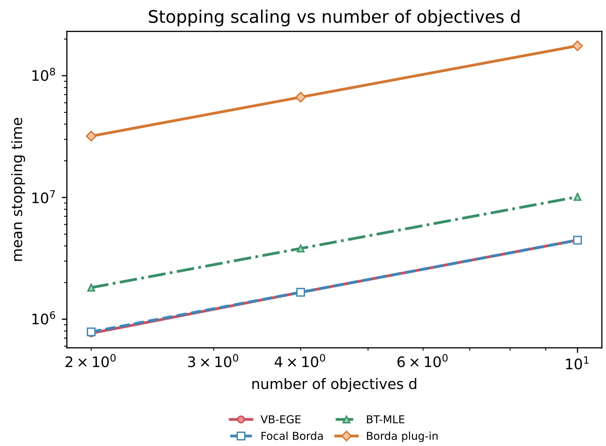
Table 9: Scaling means and baseline ratios. Parentheses contain one standard error for VB-EGE; all empirical error rates are zero.

Value	VB mean τ (SE)	Focal/VB	MLE/VB	Plug-in/VB
Gap scaling				
0.50	1.84×10^7 (3237.43)	1.00x	2.31x	39.2x
0.75	4.46×10^6 (0)	1.00x	2.27x	39.5x
1	4.46×10^6 (0)	1.00x	2.27x	39.5x
1.25	1.82×10^6 (3.17×10^4)	2.44x	1.27x	23.4x
1.50	1.07×10^6 (0)	1.00x	2.17x	39.8x

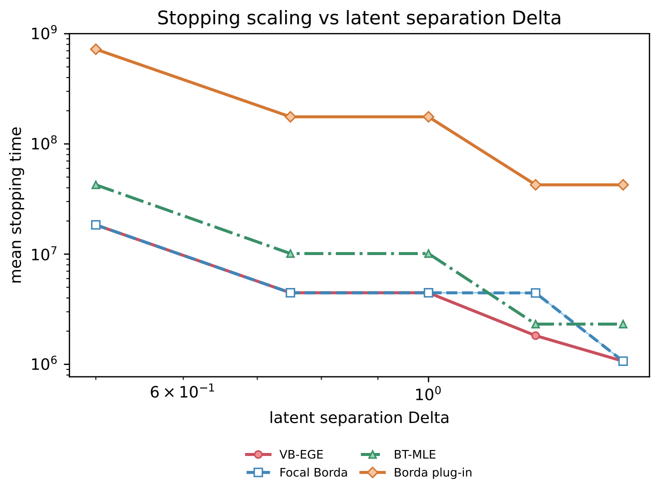
The scaling tasks isolate how stopping time changes with the number of arms, the dimension, and the latent gap on the symmetric hard instance at fixed confidence. In this generator, there is a single true Pareto arm and all other arms are dominated by the same coordinate gap. This makes the setting useful for checking whether the fixed-confidence rules behave monotonically and whether pairwise baselines pay a large cost for broad pair-coordinate coverage.



(a) K scaling

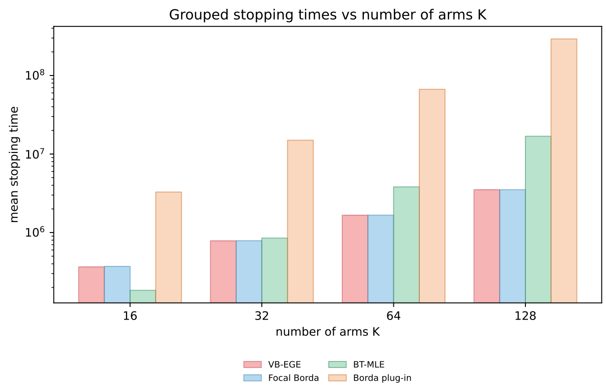


(b) d scaling

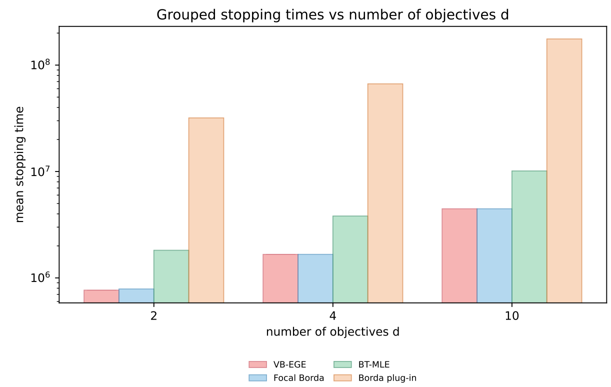


(c) Gap scaling

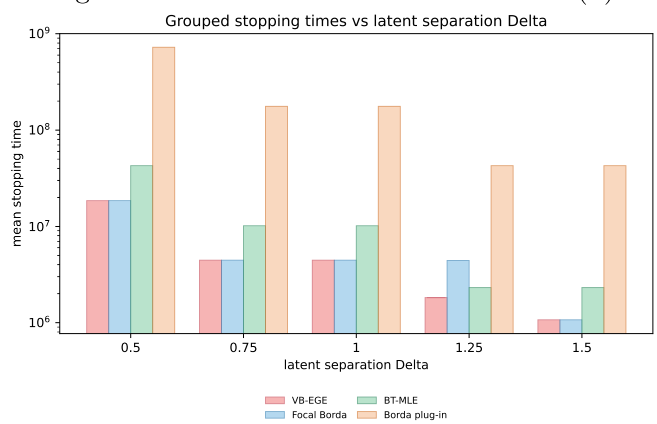
Figure 4: Scaling curves for mean stopping time. Shaded ribbons are mean $\pm$ one standard error; dark center lines show the means.



(a) K scaling



(b) d scaling



(c) Gap scaling

Figure 5: Scaling results as grouped bar plots. Darker overlays show mean $\pm$ one standard error.

Result interpretation. Descriptive log–log fits give slopes about 1.09 for both the K sweep and the d sweep, supporting the predicted near-linear leading dependence. VB-EGE-practical and UniformFocalBorda-FC are nearly tied on the symmetric hard instance, as expected because the instance has a single symmetric Pareto arm and little need for adaptive frontier structure. Both methods estimate the same coordinate-wise Vector-Borda cells, and the symmetric generator gives all dominated arms comparable gaps. Therefore most certificates mature in the same final phase, leaving little heterogeneity for VB-EGE's adaptive elimination to exploit. In this setting the near-overlap between the UniformFocalBorda-FC line and the VB-EGE-practical line is a sanity-check outcome rather than evidence against VB-EGE; the method's advantage appears when the instance has richer frontier structure or when pairwise baselines must pay broad pair-coordinate coverage costs. The pairwise BT-MLE and Borda plug-in baselines are substantially slower as K and d grow, reflecting the cost of spreading samples across many pair-coordinate cells. Larger gaps reduce stopping time. The small plateaus in the gap sweep are consistent with dyadic phase schedules: once two gap values trigger the same final phase, their stopping times can match even if the underlying gap differs.

4 Confidence-Scaling Quantiles

Definitions. The x-axis is $x = \log(1/\delta)$ and the center statistic is mean stopping time $\bar{\tau}$. Shaded ribbons and darker bar overlays are $\bar{\tau} \pm \widehat{\mathrm{SE}}(\bar{\tau})$ for that mean. The table additionally reports $q_{0.95}(\tau)$, the empirical 95th percentile, to retain tail information without adding another figure.

Table 10: Confidence-scaling quantile settings. These runs use the same fixed-confidence algorithms as the main benchmark suite and vary only the requested failure probability.

Setting	Generator	Grid	Purpose	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01, 0.005,0.002,0.001}	Stress the tail of fixed-confidence stopping times as log(1/delta) grows.	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]; delta in {0.20, 0.10,0.05,0.02,0.01,0.005}	Check confidence scaling on a high-dimensional multi-Pareto arena.	300

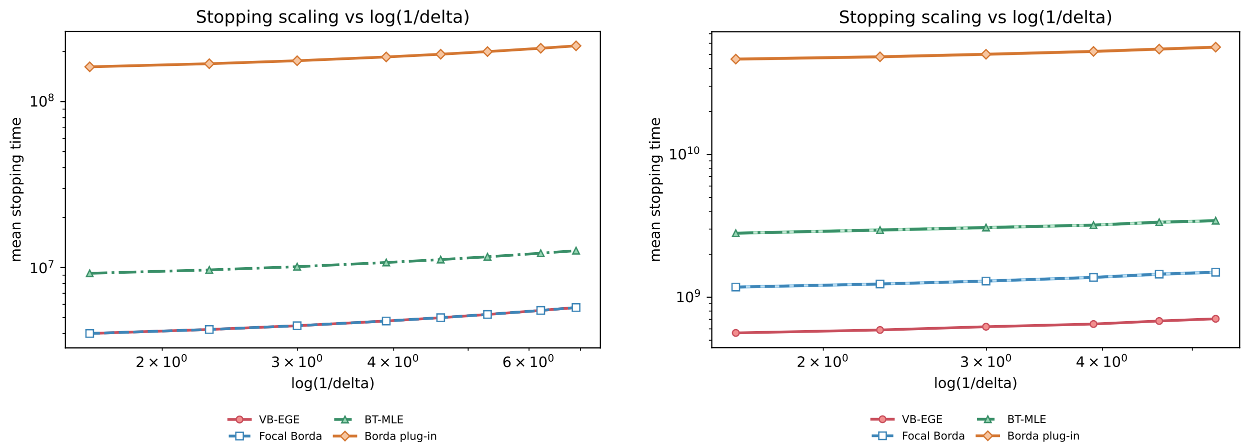
Table 11: Confidence scaling on Symmetric K64 d10. VB-EGE is reported by mean stopping time with one standard error in parentheses; q95 is retained as a tail diagnostic.

δ	VB mean τ (SE)	VB q95 τ	Focal/VB	MLE/VB	Plug-in/VB
0.200	4.01×10^6 (0)	4.01×10^6	1.00x	2.30x	40.4x
0.100	4.23×10^6 (0)	4.23×10^6	1.00x	2.29x	39.9x
0.050	4.46×10^6 (0)	4.46×10^6	1.00x	2.27x	39.5x
0.020	4.76×10^6 (0)	4.76×10^6	1.00x	2.25x	39.0x
0.010	4.99×10^6 (0)	4.99×10^6	1.00x	2.24x	38.6x
0.005	5.22×10^6 (0)	5.22×10^6	1.00x	2.23x	38.3x
0.002	5.52×10^6 (0)	5.52×10^6	1.00x	2.21x	38.0x
0.001	5.74×10^6 (0)	5.74×10^6	1.00x	2.20x	37.7x

Table 12: Confidence scaling on Arena-10 medium. VB-EGE is reported by mean stopping time with one standard error in parentheses; q95 is retained as a tail diagnostic.

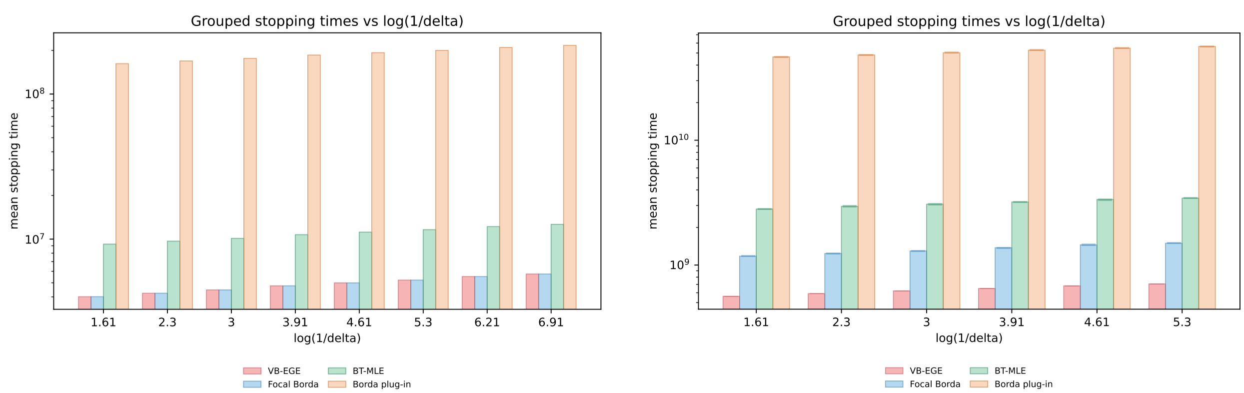
δ	VB mean τ (SE)	VB q95 τ	Focal/VB	MLE/VB	Plug-in/VB
0.200	5.62×10^8 (3.69×10^6)	6.57×10^8	2.09x	4.99x	82.7x
0.100	5.89×10^8 (3.63×10^6)	6.91×10^8	2.10x	5.01x	81.8x
0.050	6.21×10^8 (3.63×10^6)	7.10×10^8	2.09x	4.95x	80.9x
0.020	6.49×10^8 (4.46×10^6)	7.69×10^8	2.12x	4.93x	81.1x
0.010	6.81×10^8 (4.20×10^6)	7.86×10^8	2.13x	4.92x	80.2x
0.005	7.06×10^8 (4.67×10^6)	8.20×10^8	2.12x	4.87x	79.9x

This experiment repeats fixed-confidence runs over a wider grid of target failure probabilities. Within each experiment and replication, all δ values reuse the same synthetic instance, so the confidence effect is not confounded with between-instance hardness variation. The line legends are the four algorithms; the curves and bars report mean stopping time with one-standard-error bands, while the accompanying table retains tail information through q95. The x-axis is $\log(1/\delta)$, so approximately linear growth indicates the expected logarithmic confidence cost. Normalized and tau/log diagnostic plots are omitted here because they repeat the same ordering without adding a new empirical claim.



(a) Symmetric K64 d10 (b) Arena-10 medium

Figure 6: Confidence-scaling mean stopping times. Shaded ribbons are mean $\pm$ one standard error; dark center lines show the means.



(a) Symmetric K64 d10 (b) Arena-10 medium

Figure 7: Confidence-scaling grouped bar plots. Darker overlays show mean $\pm$ one standard error.

Result interpretation. On the symmetric setting, VB-EGE and the focal Borda baseline remain close because the certificate is essentially one-dimensional after symmetry. The arena setting separates the methods more clearly: VB-EGE keeps a smaller high quantile stopping time than the pairwise baselines because it does not need broad pair-coordinate coverage at every confidence level. The final-phase diagnostic should move in steps rather than smoothly, which is expected from phase doubling. A direct fit $\bar{\tau} = a + b \log(1/\delta)$ gives $R^2 \approx 0.99997$ on the symmetric task and $R^2 \approx 0.9978$ on Arena-10. This supports logarithmic confidence cost in stopping time; it does not empirically validate a 0.001 error probability from only 500 replications.

5 Fixed-Confidence Benchmark Suite

Definitions. Each benchmark bar reports mean stopping time $\bar{\tau}$ on a log scale. The darker interval overlay spans $\bar{\tau} \pm \widehat{\mathrm{SE}}(\bar{\tau})$. Baseline ratios in the table are paired per-run mean ratios to VB-EGE with cluster-aware standard errors. Lower bars and ratios above one favor VB-EGE.

Table 13: Fixed-confidence benchmark settings. All algorithms use target failure probability $\delta = 0.05$; $|P|$ denotes the true Pareto set size.

Setting	Generator	K	d	$\lvert P\rvert$	Reps	Parameters
Convex-2D	convex_frontier_2d	60	2	15	500	s=15
Convex-3D	convex_frontier_3d	60	3	15	500	s=15, simplex alpha=1, margins [0.03,0.18]
Arena-4 small	arena_tradeoff_frontier	32	4	8	500	s=8, margins [0.08,0.25], alpha=0.7
Arena-4 medium	arena_tradeoff_frontier	64	4	12	300	s=12, margins [0.08,0.25], alpha=0.7
Arena-10 medium	arena_tradeoff_frontier	64	10	16	300	s=16, margins [0.06,0.20], alpha=0.7
Witness-4	unique_witness_d	40	4	8	500	s=8, q=4, margins [0.04,0.16]
Witness-10	unique_witness_d	80	10	16	300	s=16, q=4, margins [0.035,0.14]
Two-group-10	highdim_two_group	64	10	15.79	300	K_low=48, K_high=16

Table 14: Benchmark mean stopping times and paired mean slowdown ratios: Convex and witness settings. Parentheses contain one cluster-aware standard error. All empirical error rates are zero.

Setting	VB mean τ (SE)	Focal/VB (SE)	MLE/VB (SE)	Plug-in/VB (SE)
Convex-2D	1.21×10^8 (9.57×10^5)	3.46x (0.10x)	6.74x (0.21x)	115x (3.50x)
Convex-3D	4.46×10^9 (3.86×10^9)	9.85x (1.08x)	21.2x (2.36x)	341x (38.8x)
Witness-4	2.28×10^8 (7.27×10^5)	1.26x (0.00x)	1.80x (0.01x)	29.8x (0.10x)
Witness-10	1.61×10^9 (4.76×10^6)	2.15x (0.08x)	2.96x (0.01x)	48.8x (0.13x)

Table 15: Benchmark mean stopping times and paired mean slowdown ratios: Arena and two-group settings. Parentheses contain one cluster-aware standard error. All empirical error rates are zero.

Setting	VB mean τ (SE)	Focal/VB (SE)	MLE/VB (SE)	Plug-in/VB (SE)
Arena-4 small	3.83×10^{10} (3.30×10^{10})	9.46x (0.36x)	10.1x (0.38x)	170x (6.15x)
Arena-4 medium	3.65×10^{10} (1.60×10^{10})	19.8x (0.87x)	42.9x (1.51x)	784x (37.2x)
Arena-10 medium	6.13×10^8 (6.46×10^6)	2.08x (0.02x)	4.88x (0.06x)	80.2x (0.88x)
Two-group-10	7.16×10^{11} (6.15×10^{11})	17.7x (0.85x)	39.9x (1.71x)	665x (27.6x)

The benchmark suite stresses different Pareto geometries beyond the symmetric hard case. Convex-2D and Convex-3D test clean low-dimensional trade-off frontiers. Arena settings use mutually non-dominated anchors and dominated alternatives offset from a random anchor. Witness settings create dominated arms that are certified by a unique Pareto witness, which tests whether algorithms can avoid unnecessary global pairwise coverage. Two-group-10 is a high-dimensional separation task with many low-quality arms and a smaller high-quality group.

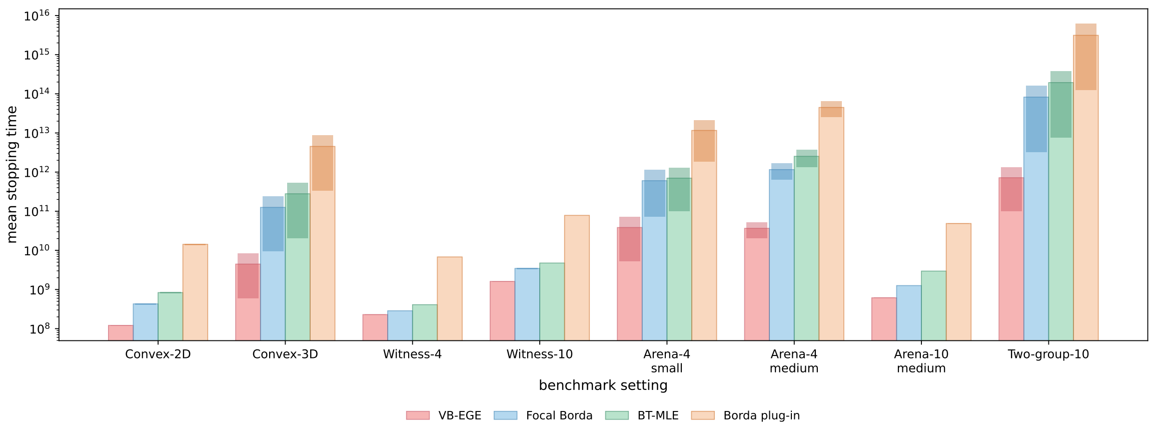


Figure 8: Stopping times across all eight fixed-confidence benchmark settings. Darker overlays show mean ± one standard error.

Result interpretation. VB-EGE-practical has the smallest mean stopping time in every benchmark setting while maintaining zero empirical error. The advantage is moderate on unique-witness instances, where the focal Borda baseline already receives local certificate information, and much larger on arena and two-group instances. The BT-MLE baseline has near-perfect pairwise sign accuracy in the completed runs, but it pays a large exploration cost before enough pair-coordinate cells are covered. The Borda plug-in pairwise baseline is consistently the slowest, which suggests that uniform pairwise sampling is poorly matched to fixed-confidence Pareto certification when the certificate can be expressed in the Vector-Borda embedding.

6 Pareto-Size Ablation

Definitions. The x-axis is the true Pareto-front size $|P| = s$. Each dark line is mean stopping time $\bar{\tau}$, and its translucent ribbon spans $\bar{\tau} \pm \widehat{\mathrm{SE}}(\bar{\tau})$ across replications. The result table reports ratios of mean stopping times to VB-EGE at the same (d, s) cell; ratios above one favor VB-EGE.

Table 16: Pareto-size ablation settings. The arena generator is held fixed while the target frontier size $s = |P|$ changes.

Setting	Generator	Grid	Purpose	Reps
Arena-4	arena_tradeoff_frontier	K=64; d=4; s in {4,8,16,32}; margins [0.08,0.25]	Measure how certificate cost changes as the frontier grows.	300
Arena-10	arena_tradeoff_frontier	K=64; d=10; s in {4,8,16,32}; margins [0.06,0.20]	Measure how certificate cost changes as the frontier grows.	300

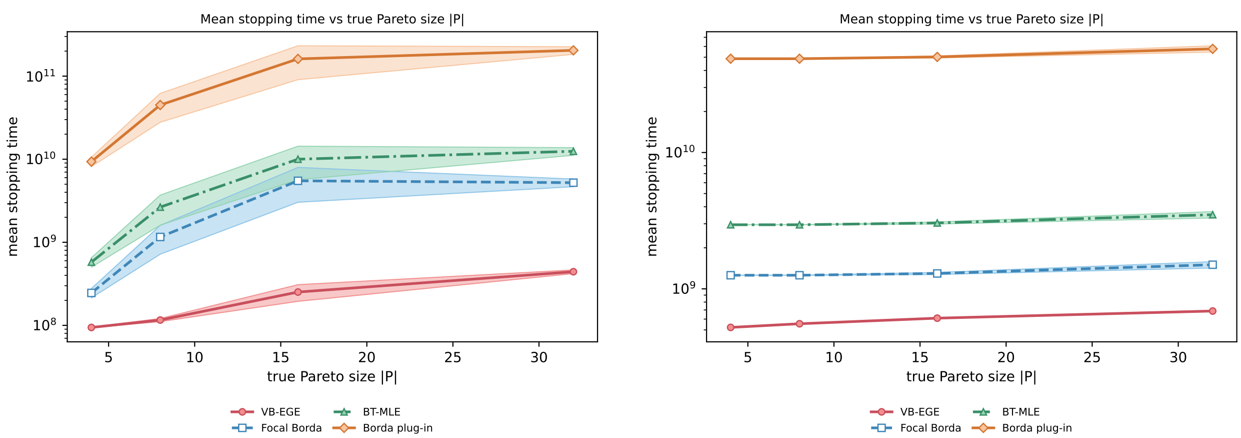
Table 17: Pareto-size means on Arena-4. Parentheses contain one standard error; ratios compare mean stopping times at the same target $|P|$.

$\|P\|$	VB mean τ (SE)	Focal/VB	MLE/VB	Plug-in/VB
4	9.44×10^7 (7.02×10^5)	2.60x	6.11x	99.1x
8	1.16×10^8 (4.91×10^6)	10.00x	22.9x	387x
16	2.52×10^8 (5.70×10^7)	21.8x	39.7x	640x
32	4.42×10^8 (2.22×10^7)	11.8x	28.2x	462x

Table 18: Pareto-size means on Arena-10. Parentheses contain one standard error; ratios compare mean stopping times at the same target $|P|$.

$\|P\|$	VB mean τ (SE)	Focal/VB	MLE/VB	Plug-in/VB
4	5.21×10^8 (2.79×10^6)	2.41x	5.66x	93.5x
8	5.55×10^8 (3.11×10^6)	2.27x	5.32x	87.9x
16	6.10×10^8 (4.21×10^6)	2.13x	4.99x	82.4x
32	6.88×10^8 (6.87×10^6)	2.19x	5.09x	83.7x

This ablation holds the arena geometry fixed while increasing the true Pareto frontier size $|P|$. Arena-4 and Arena-10 retain the same generator families and margin ranges while the requested number of frontier arms changes.



(a) Arena-4 tau

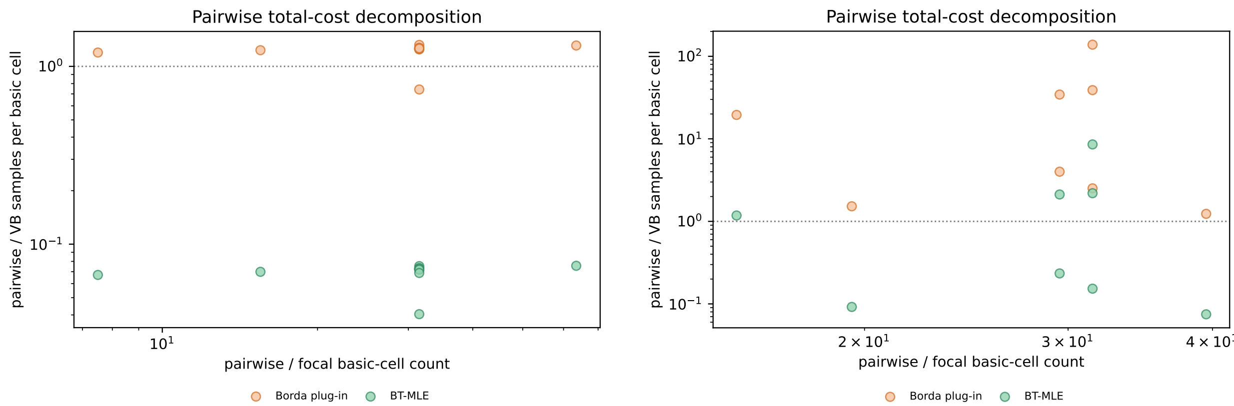
(b) Arena-10 tau

Figure 9: Pareto-size stopping-time ablation. Shaded ribbons are mean ± one standard error; dark center lines show the means.

Result interpretation. The important question is whether the relative advantage of VB-EGE persists as the Pareto frontier grows. When the ratio curves stay above one, the adaptive Vector-Borda certificate remains more sample-efficient than uniform baselines even though more arms need to be accepted.

7 Sanity Check: Sampling-Mechanism Diagnostics

Definitions. For pairwise methods, $C = dK(K-1)/2$ is the number of pair-coordinate cells; VB-EGE uses Kd focal arm-coordinate cells. The structural multiplier is $C/(Kd) = (K-1)/2$. The per-cell burden ratio is $(\tau_{\mathrm{pair}}/C)/(\tau_{\mathrm{VB}}/(Kd))$. The allocation diagnostic plots each arm's Borda certificate gap Δ_i^B against its total VB-EGE allocation $\sum_r N_{ir}$.



(a) Scaling tasks

(b) Benchmark tasks

Figure 10: Pairwise total-cost decomposition. Their product is the total stopping-time ratio $\tau_{\mathrm{pair}}/\tau_{\mathrm{VB}}$.

This decomposition separates the number of basic cells from the average evidence collected in each cell. Pairwise methods pay for both a larger cell set and, depending on the certificate, a different per-cell burden.

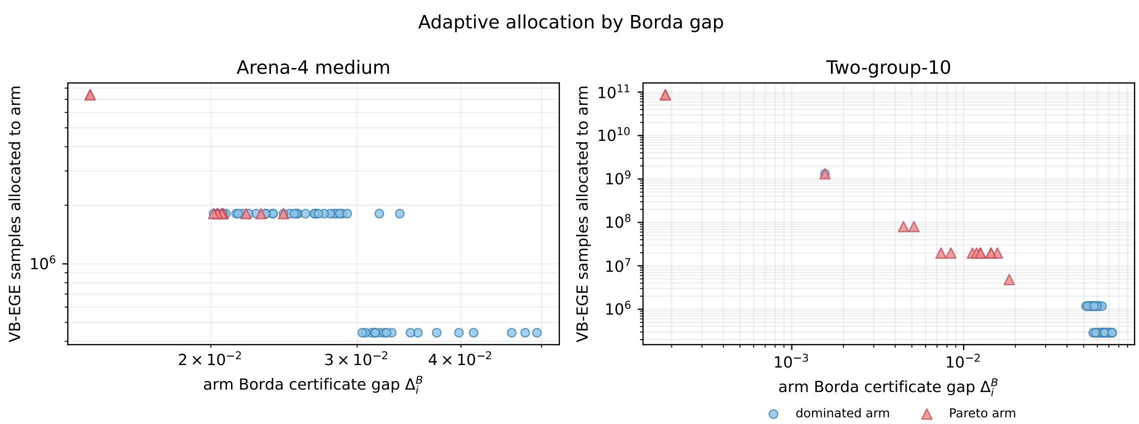


Figure 11: Arm allocation versus Borda certificate gap. Smaller-gap arms receive more VB-EGE samples in the representative runs.

The allocation plot is the direct adaptivity diagnostic: VB-EGE concentrates its sampling on arms close to the decision boundary instead of maintaining uniform coverage over all arms or all pair-coordinate cells.

8 Takeaways

1. All completed main fixed-confidence synthetic experiments achieved zero empirical Pareto-set error.
2. VB-EGE-practical loses essentially nothing to uniform focal Borda on symmetric instances and is substantially more sample-efficient on heterogeneous Pareto geometries.
3. Pairwise BT-MLE can estimate latent pairwise signs accurately, but fixed-confidence Pareto certification makes its broad pair-coordinate coverage expensive.
4. The largest gains occur in high-dimensional arena and two-group settings, where adaptive Vector-Borda certificates avoid many unnecessary pairwise comparisons.
5. Arena-10 benchmark and constant-sensitivity results now use one versioned instance bank and agree exactly at the common default configuration; confidence scaling retains a separate paired-δ bank.
6. The BT-MLE comparator is an empirical fixed-confidence-style certificate rather than a method with a formal delta-correct confidence region.

Reproducibility Notes

The source summaries used in this report are `results/summary/fixed_confidence_scaling_summary.csv` and `results/summary/fixed_confidence_benchmarks_summary.csv`, plus `results/summary/fixed_confidence_benchmarks_summary_paired.csv`, `results/summary/confidence_scaling_quantile_summary.csv`, `results/summary/constants_calibration_summary.csv`, `results/summary/pareto_size_ablation_summary.csv`, and `results/summary/theory_constants_sanity_summary.csv`. The raw completed main outputs are `results/raw/fixed_confidence_scaling.csv`, `results/raw/fixed_confidence_benchmarks.csv`, and the corresponding CSV fallbacks under `results/raw/` for the diagnostic extensions. Smoke runs are excluded from the main conclusions because they were used only to validate the implementation path before the full runs. Revised randomized benchmarks store `instance_id`, `theta_hash`, instance index, and observation replicate in every raw row.